

(3.13) Let p be a prime number, $n \geq 3$ an integer prime to p , and $F_{n,p}$ the functor which associates to each scheme S the set of isomorphism classes of commutative diagrams of S -schemes

$$\begin{array}{ccc}
 & \mathbb{Z}/(n)^2 & \\
 \alpha \nearrow & & \nwarrow \alpha' \\
 E_n & \longrightarrow & F_n \\
 \cap & & \cap \\
 E & \xrightarrow{\varphi} & F
 \end{array} \tag{3.14}$$

where ϕ is a p -isogeny between elliptic curves and α an isomorphism. One denotes by q_1 and $q_2: F_{n,p} \rightarrow M_n$ the morphisms of functors associating to a diagram (3.14) the subdiagrams (E, E_n, α) and (F, F_n, α') .

Proposition 3.15 *The functor $F_{n,p}$ is represented by a scheme $M_{n,p}$, and the morphisms $q_1, q_2: M_{n,p} \rightarrow M_n$ are finite.*

The automorphism σ of $F_{n,p}$ sending $\varphi: E \rightarrow F$ to ${}^t\varphi: F \rightarrow E$ exchanges q_1 and q_2 ; it suffices thus to consider q_1 . This morphism identifies $F_{n,p}$ with the functor of subgroups of order p of the universal elliptic curve E over M_n , so that, by the theory of Hilbert schemes, $F_{n,p}$ is representable and $M_{n,p}$ is proper over M_n . If s is a geometric point of M_n , $q_1^{-1}(s)$ is the set of subgroups of order p of E_s , and has $p+1$ elements if $\text{char}(k(s)) \neq p$, and only one (the kernel of Frobenius) if $\text{char}(k(s)) = p$. $\square$

One can show that $M_{n,p}$ is regular, and that q_1 and q_2 are finite flat; we don't use this delicate result, but we content ourselves here to note that over $\text{Spec}(\mathbb{Z}[1/p])$, each q_i makes $M_{n,p}$ an étale covering of degree $p+1$ of M_n .

The morphisms q_i can be inserted into a commutative diagram

$$\begin{array}{ccccc}
 & q_1^* E & \xrightarrow{\varphi} & q_2^* E & \\
 & \swarrow & u \searrow & \swarrow v & \searrow \\
 E & & M_{n,p} & & E \\
 & f_n \searrow & \swarrow q_1 & q_2 \searrow & \swarrow f_n \\
 & M_n & & M_n &
 \end{array} \tag{3.16}$$

where (φ, u, v) is a part of the universal diagram (3.14).

One denotes by I_p the morphism from M_n to M_n corresponding to the morphism of functors $(E, \alpha) \mapsto (E, \alpha/p)$:

$$\begin{array}{ccc}
 E & \longrightarrow & E \\
 \downarrow & & \downarrow \\
 M_n & \xrightarrow{I_p} & M_n
 \end{array} \quad I_p^*(E, \alpha) = (E, \alpha/p). \tag{3.17}$$

I_p^* is an automorphism of $\tilde{H}^i(M_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}))$.

It is tiresome, but routine, to prove

Proposition 3.18 (i) The endomorphism T_p of ${}_nW$ is expressed, with the aid of (3.16), as the composite map

$$\begin{aligned} \tilde{H}^1(M_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Q}})) &\xrightarrow{q_2^*} \tilde{H}^1(M_{n,p}^{\text{an}}, \text{Sym}^k(R^1 v_* \underline{\mathbb{Q}})) \\ &\xrightarrow{\varphi^*} \tilde{H}^1(M_{n,p}^{\text{an}}, \text{Sym}^k(R^1 u_* \underline{\mathbb{Q}})) \xrightarrow{q_{1*}} \tilde{H}^1(M_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Q}})), \end{aligned}$$

where q_{1*} is the “trace morphism” for the covering q_1 .

(ii) Similarly, $R_p = p^k I_p^*$. $\square$

The distrustful reader could forget the adelic preliminaries and define T_p by (i).

When $n = 1$ or 2 , one puts ${}_nW = W^{K_n}$, so that

$${}_1W = {}_nW^{GL_2(\mathbb{Z}/(n))}.$$

If S_{k+2} denotes the space of cusp forms, for the group $SL_2(\mathbb{Z})$, of weight $k + 2$, the Shimura isomorphism (3.11) induces an isomorphism

$${}_1^k W_\infty = {}_1^k W \otimes \mathbb{C} = S_{k+2} \oplus \overline{S_{k+2}}.$$

It is tiresome, but routine, to prove

Proposition 3.19 The endomorphism T_p of ${}_1^k W_\infty$ is identified, via the Shimura isomorphism, with the direct sum of the Hecke operator on S_{k+2} (including the factor p^{k-1}), and its conjugate. $\square$

(3.20) One has canonically

$$\bigwedge^2 R^1 f_{n*} \underline{\mathbb{Z}}_\ell \sim R^2 f_{n*} \underline{\mathbb{Z}}_\ell \sim \underline{\mathbb{Z}}_\ell(-1),$$

so that $\text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}}_\ell)$ is equipped with a bilinear form (symmetric for k even, alternating for k odd) with values in $\underline{\mathbb{Z}}_\ell(-k)$. The form induced by tensoring with $\mathbb{Q}_\ell$ is nondegenerate.

If $\underline{F}$ is a lcc $\mathbb{Q}_\ell$ -sheaf over a smooth scheme X of pure dimension n over an algebraically closed field k , then Poincaré duality gives

$$\begin{aligned} H^i(X, \underline{F})^\vee &\sim H_c^{2n-i}(X, \underline{\text{Hom}}(\underline{F}, \underline{\mathbb{Q}}_\ell(n))) \\ H_c^i(X, \underline{F})^\vee &\sim H^{2n-i}(X, \underline{\text{Hom}}(\underline{F}, \underline{\mathbb{Q}}_\ell(n))) \\ \tilde{H}^i(X, \underline{F})^\vee &\sim \tilde{H}^{2n-i}(X, \underline{\text{Hom}}(\underline{F}, \underline{\mathbb{Q}}_\ell(n))). \end{aligned}$$

Taking $X = \overline{M}_n$ and $\underline{F} = \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Q}}_\ell)$ in these considerations, one defines a nondegenerate bilinear form ${}_n(\ , \)$ on ${}_n^k W_\ell$ with values in $\mathbb{Q}_\ell(-k-1)$. This form is symmetric for k odd, and alternating for k even. This is the ℓ -adic analogue of the Peterson inner product. If $n \mid m$, and if the covering $\psi: M_m \rightarrow M_n$ is of degree d , then one has

$${}_m(\psi^* x, \psi^* y) = d \cdot {}_n(x, y).$$

4 The Congruence Formula.

One fixes in this n° integers $k \geq 0$ and $n \geq 3$, and prime numbers p and ℓ . One supposes that p is prime to ℓ and n . One denotes by $f_n: E \rightarrow M_n$ the universal elliptic curve over M_n , equipped with $\alpha: E_n \xrightarrow{\sim} \mathbb{Z}/(n)^2$.

Whatever the scheme Y , one denotes by a the unique morphism of Y to $\text{Spec}(\mathbb{Z})$, or, if necessary, to a subscheme of $\text{Spec}(\mathbb{Z})$. If Y is separated and of finite type over $\text{Spec}(\mathbb{Z})$, and if $\underline{F}$ is a $\mathbb{Z}_\ell$ - or $\mathbb{Q}_\ell$ -sheaf on Y , one denotes by $R^i a_*(Y, \underline{F})$ (resp. $R^i a_!(Y, \underline{F})$, resp. $R^i \hat{a}(Y, \underline{F})$) the $\mathbb{Z}_\ell$ - or $\mathbb{Q}_\ell$ -sheaf over $\text{Spec}(\mathbb{Z})$ that is the i th higher direct image of $\underline{F}$ under a (resp. i th direct image with proper supports, resp. $\text{Im}(R^i a_!(Y, \underline{F}) \rightarrow R^i a_*(Y, \underline{F}))$). One puts, for $m \in \mathbb{N}$, $Y[1/m] = Y \times \text{Spec}(\mathbb{Z}[1/m])$.

Theorem 4.1 (Igusa [1]) *The scheme M_n can be compactified to a scheme of curves M_n^* that is projective and smooth over $\text{Spec}(\mathbb{Z}[1/n])$, such that $M_n^* \setminus M_n$ is an étale covering of $\text{Spec}(\mathbb{Z}[1/n])$.*

The schemes M_n is formally smooth, thus smooth over $\text{Spec}(\mathbb{Z})$.

The modular invariant j of the universal curve over M_n is a morphism of M_n to the affine line A^1 over $\text{Spec}(\mathbb{Z}[1/n])$. The morphism j is finite, and is an étale covering away from the sections 0 and 1728 of A^1 ; in fact:

(a) Two elliptic curves over an algebraically closed field with the same j -invariant are isomorphic (e.g., [8] 6.3), thus the geometric fibers of j are finite. The schemes M_n and A^1 being smooth of the same dimension over $\text{Spec}(\mathbb{Z})$, j is quasi-finite and flat.

(b) If E is an elliptic curve over the field of fractions K of a discrete valuation ring R , of j -invariant in R , and whose points of order n are rational over K , then E has a good reduction. The valuative criterion for properness shows thus that j is proper.

(c) If E and F are two elliptic curves over a scheme S , of the same j -invariant, and if j and $j - 1728$ are invertible, then the scheme $\underline{\text{Isom}}(S; E, F)$ of isomorphisms between E and F is étale over S ([8] 6.3). In the diagram

$$\begin{array}{ccc} \underline{\text{Isom}}(M_n \times_{A^1} M_n; \text{pr}_1^* E, \text{pr}_2^* E) & \sim & M_n \times GL_2(\mathbb{Z}/(n)) \\ \downarrow u & & \downarrow v \\ M_n \times_{A^1} M_n & \xrightarrow{\text{pr}_1} & M_n \end{array},$$

where $j \neq 0, 1728$, u and v are étale surjective, thus pr_1 is étale and, by flat descent, j is étale.

The section at infinity of the projective line $\mathbb{P}^1 \supset A^1$ over $\text{Spec}(\mathbb{Z}[1/n])$ is a regular divisor, whose generic point is of characteristic 0, in a regular scheme. It results then from a theorem of Abyankhar (see [5]) that along this divisor $j = \infty$, M_n is moderately (tamely?) ramified over $\mathbb{P}^1$, and that the normalization M_n^* of $\mathbb{P}^1$ in M_n satisfies (4.1). $\square$

It results from the same theorem that the lcc $\mathbb{Z}_\ell$ -sheaves $R^i f_{n*} \underline{\mathbb{Z}_\ell}$ on $M_n[1/\ell]$ are moderately ramified at infinity. Hence, from (4.1) and the specialization theorems for ℓ -adic cohomology (see [5]), it results that $R^i a_*(M_n, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}_\ell}))$, $R^i a_!(M_n, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}_\ell}))$, and thus $R^i \hat{a}(M_n, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}_\ell}))$ are lcc $\mathbb{Z}_\ell$ -sheaves over $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$, whose formation is compatible with all base changes.

Corollary 4.2 *The Galois module ${}_nW_\ell$ is the fiber of the lcc $\mathbb{Q}_\ell$ -sheaf*

$$R^i\tilde{a}(M_n, \text{Sym}^k(R^1f_{n*}\underline{\mathbb{Z}}_\ell)) \otimes \mathbb{Q}_\ell$$

over $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$ at the geometric point $\overline{\mathbb{Q}}$. It is unramified away from n and ℓ . $\square$

Let the two commutative diagrams over $M_n \otimes \mathbb{F}_p$

$$\begin{array}{ccc} & \mathbb{Z}/(n)^2 & \\ \alpha \nearrow & & \nwarrow \alpha^{(p)} \\ E_n & \longrightarrow & E_n^{(p)} \\ \cap & & \cap \\ E & \xrightarrow{F} & E^{(p)} \end{array} \quad \text{and} \quad \begin{array}{ccc} & \mathbb{Z}/(n)^2 & \\ p\alpha^{(p)} \nearrow & & \nwarrow \alpha \\ E_n^{(p)} & \longrightarrow & E_n \\ \cap & & \cap \\ E^{(p)} & \xrightarrow{V} & E \end{array},$$

be written more briefly

$$F: (E, \alpha) \rightarrow (E^{(p)}, \alpha^{(p)}) \quad \text{and} \quad V: (E^{(p)}, p\alpha^{(p)}) \rightarrow (E, \alpha),$$

where F is the Frobenius morphism and V , its transpose, is the “Verschiebung.” These diagrams define morphism Φ_1 and Φ_2 of $M_n \otimes \mathbb{F}_p$ to $M_{n,p}$. These morphisms are finite, considered as sections of q_1 and q_2 , and define a morphism

$$\Phi = \Phi_1 \amalg \Phi_2: M_n \otimes \mathbb{F}_p \amalg M_n \otimes \mathbb{F}_p \rightarrow M_{n,p} \otimes \mathbb{F}_p.$$

Let Φ^h be the restriction of Φ to the opens M_n^h and $M_{n,p}^h$ of $M_n \otimes \mathbb{F}_p$ and $M_{n,p} \otimes \mathbb{F}_p$ corresponding to curves of nonzero Hasse invariant h .

Proposition 4.3 Φ^h is an isomorphism.

Let $\varphi: E_1 \rightarrow E_2$ be a p -isogeny between elliptic curves of invertible Hasse invariant over a scheme S of characteristic p . At each geometric point of S , either the kernel $\ker(\varphi)$ of φ is étale over S , or its Cartier dual, isomorphic to $\ker({}^t\varphi)$, is étale over S . The property “ $\ker(\varphi)$ is étale” is an open property, so that, locally on S , either $\ker(\varphi)$ is purely infinitesimal, or $\ker({}^t\varphi)$ is. The only infinitesimal subgroup of order p of E_1 or E_2 being the kernel of Frobenius, in the first case, φ is isomorphic to $F: E_1 \rightarrow E_1^{(p)}$, and in the second case, ${}^t\varphi$ is isomorphic to $F: E_2 \rightarrow E_2^{(p)}$ and φ to $V: E_2^{(p)} \rightarrow E_2$. $\square$

Proposition 4.4 (i) *The scheme $M_{n,p}$ is smooth over $\text{Spec}(\mathbb{Z})$ away from points of characteristic p where $h = 0$.*

(ii) *The morphisms q_1 and q_2 induce finite flat morphisms q'_1 and q'_2 from the normalization $M'_{n,p}$ of $M_{n,p}$ to M_n .*

(iii) *The morphism Φ can be factored through a surjective morphism*

$$\Phi': M_n \otimes \mathbb{F}_p \amalg M_n \otimes \mathbb{F}_p \rightarrow M'_{n,p} \otimes \mathbb{F}_p.$$

The automorphism σ of (3.15) exchanges φ and ${}^t\varphi$, so that it suffices to prove (i) at the points of characteristic p of $M_{n,p}$ where the kernel of φ is infinitesimal: there is no obstruction to infinitesimally lifting an elliptic curve and the infinitesimal part of the kernel of multiplication by p .

Where $p = h = 0$, the fiber of the finite morphism (3.15) $q_i: M_{n,p} \rightarrow M_n$ is reduced to a point, so that the open of smoothness of $M_{n,p}$ is dense in $M_{n,p}$ and that $M'_{n,p}$ is everywhere of dimension 2. The scheme M_n being regular, following EGA 0_{IV} 16.5.1 and 17.3.5(ii), the morphism $q_i: M'_{n,p} \rightarrow M_n$ is flat. Finally, (iii) results from the fact that Φ is finite and $M_n \otimes \mathbb{F}_p$ is a normal curve. $\square$

The Hecke endomorphism T_p of ${}_nW_\ell$, such as is made explicit in (3.18), is the tensorization with $\mathbb{Q}_\ell$ of the fiber at the geometric point $\overline{\mathbb{Q}}$ of $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$ of the endomorphism (again denoted by T_p) of $R^1\tilde{a}(M_n, \text{Sym}^k(R^1f_{n*}(\mathbb{Z}_\ell)))$ defined by the “correspondence”

$$\begin{array}{ccccc}
 & q_1^*E & \xrightarrow{\varphi} & q_2^*E & \\
 \swarrow & & \searrow u & & \swarrow v \\
 E & & M'_{n,p} & & E \\
 \searrow f_n & \swarrow q'_1 & & \searrow q'_2 & \swarrow f_n \\
 & M_n & & M_n & \\
 & T_p = q_{1*}\varphi^*q_2'^* & \text{(cf. 3.18)} & &
 \end{array} \quad (4.5)$$

The endomorphisms R_p and I_p can be interpreted in the same way.

Lemma 4.6 *Let X , Y , Z_1 , and Z_2 be four schemes that are separated and of finite type over a noetherian scheme S , let $\underline{F}$ be a $\mathbb{Z}_\ell$ -sheaf on X , $\underline{G}$ a $\mathbb{Z}_\ell$ -sheaf on Y , and denote by a each of the structural maps from X , Y , Z_1 or Z_2 to S . Suppose we have the following commutative diagram of S -schemes and morphisms of sheaves:*

$$\begin{array}{ccc}
 y_1^*\underline{G} & \xrightarrow{z_1} & x_1^*\underline{F} \\
 Z_1 & \xrightarrow{f} & Z_2 \\
 x_1 \downarrow \swarrow x_2 & & y_1 \searrow \downarrow y_2 \\
 X & & Y
 \end{array}$$

*Suppose that $f^*z_2 = z_1$, that y_1 and y_2 are proper, that x_1 and x_2 are finite and flat, and that, for every geometric point s of Z_2 , the multiplicity of s in its fiber $x_2^{-1}(x_2(s))$ is equal to the sum of the multiplicities in their fibers (for x_1) of the geometric points of Z_1 in the inverse image of s under f . Then, the diagram*

$$\begin{array}{ccccccc}
 R^i\tilde{a}(Y, \underline{G}) & \xrightarrow{y_1^*} & R^i\tilde{a}(Z_1, \underline{G}) & \xrightarrow{z_1} & R^i\tilde{a}(Z_1, \underline{F}) & \xrightarrow{x_1^*} & R^i\tilde{a}(X, \underline{F}) \\
 \parallel & & f^* \uparrow & & f^* \uparrow & & \parallel \\
 R^i\tilde{a}(Y, \underline{G}) & \xrightarrow{y_2^*} & R^i\tilde{a}(Z_2, \underline{G}) & \xrightarrow{z_2} & R^i\tilde{a}(Z_2, \underline{F}) & \xrightarrow{x_2^*} & R^i\tilde{a}(X, \underline{F})
 \end{array}$$

commutes.

This lemma results from the analogous lemmas for $R^i a_!$ and $R^i a_*$. The commutativity of the first squares is trivial. The last is rewritten

$$\begin{array}{ccccc}
 R^i\tilde{a}(Z_1, x_1^*\underline{F}) & \xleftarrow{\sim} & R^i\tilde{a}(X, x_{1*}x_1^*\underline{F}) & \xrightarrow{\text{Tr}} & R^i\tilde{a}(X, \underline{F}) \\
 \uparrow & & \uparrow & & \parallel \\
 R^i\tilde{a}(Z_2, x_2^*\underline{F}) & \xleftarrow{\sim} & R^i\tilde{a}(X, x_{2*}x_2^*\underline{F}) & \xrightarrow{\text{Tr}} & R^i\tilde{a}(X, \underline{F})
 \end{array}$$

and one recalls the definition of the trace in verifying that the square

$$\begin{array}{ccc} x_{1*}x_1^*F & \xrightarrow{\text{Tr}} & \underline{F} \\ \uparrow & & \parallel \\ x_{2*}x_2^*F & \xrightarrow{\text{Tr}} & \underline{F} \end{array}$$

is commutative. $\square$

(4.7) One denotes by $T_p/\mathbb{F}_p$ the endomorphism induced by T_p on the restriction to $\text{Spec}(\mathbb{F}_p)$ of the lcc $\mathbb{Z}_\ell$ -sheaf $R^1\tilde{a}(M_n, \text{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell)$. One has

$$R^1\tilde{a}(M_n, \text{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell) \mid \text{Spec}(\mathbb{F}_p) \xrightarrow{\sim} R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell);$$

the formation of the trace morphism for a finite flat morphism is compatible with change of base, so that $T_p/\mathbb{F}_p$ can be constructed, over the model (3.18), starting from the *fiber* over $\mathbb{F}_p$ of the “correspondence” (4.5). Lemma (4.6), applied to the commutative diagrams

$$\begin{array}{ccc} M_n \otimes \mathbb{F}_p \amalg M_n \otimes \mathbb{F}_p & \xrightarrow{\Phi'} & M'_{n,p} \otimes \mathbb{F}_p \\ \downarrow & \swarrow q'_1 & \\ M_n \otimes \mathbb{F}_p & & \end{array} \quad \begin{array}{ccc} M_n \otimes \mathbb{F}_p \amalg M_n \otimes \mathbb{F}_p & \xrightarrow{\Phi'} & M'_{n,p} \otimes \mathbb{F}_p \\ q'_2 \searrow & & \downarrow \\ & & M_n \otimes \mathbb{F}_p, \end{array}$$

furnishes then a decomposition of $T_p/\mathbb{F}_p$ into the sum of endomorphisms defined by the two following correspondences:

$$(a) \quad \begin{array}{ccccc} & (E, \alpha) & \xrightarrow{F} & (E^{(p)}, \alpha^{(p)}) = F^*(E, \alpha) & \\ & \parallel & \searrow & \swarrow & \\ (E, \alpha) & & M_n \otimes \mathbb{F}_p & & (E, \alpha) \\ & \searrow & \parallel & \searrow F & \\ & M_n \otimes \mathbb{F}_p & & M_n \otimes \mathbb{F}_p & \end{array}$$

where F is the absolute Frobenius. One recognizes in this correspondence the *geometric* Frobenius

$$(b) \quad \begin{array}{ccccc} & x^*(E, \alpha) = (E^{(p)}, p\alpha^{(p)}) & \xrightarrow{V} & (E, \alpha) & \\ & \swarrow & f_n^{(p)} \searrow & \swarrow f_n & \parallel \\ (E, \alpha) & & M_n \otimes \mathbb{F}_p & & (E, \alpha) \\ & \searrow & \swarrow x & \searrow & \\ & M_n \otimes \mathbb{F}_p & & M_n \otimes \mathbb{F}_p & \end{array}$$

The map x is the composite map $I_p^{-1} \circ F$:

$$\begin{array}{ccccc} (E, \alpha) & \longleftarrow & (E, p\alpha) & \longleftarrow & (E^{(p)}, p\alpha^{(p)}) \\ \downarrow & & \downarrow & & \downarrow \\ M_n \otimes \mathbb{F}_p & \xleftarrow{I_p^{-1}} & M_n \otimes \mathbb{F}_p & \xleftarrow{F} & M_n \otimes \mathbb{F}_p \end{array}.$$

The corresponding endomorphism is thus the composite of

$$\begin{aligned} V &: R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \xrightarrow{V^*} R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*}^{(p)}\mathbb{Z}_\ell)) \\ &\xrightarrow{\text{Tr}_F} R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \end{aligned}$$

and of

$$I_p^* = \text{Tr}_{I_p^{-1}} : \text{endomorphism of } R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k R^1 f_{n*} \underline{\mathbb{Z}}_\ell).$$

Proposition 4.8 *One has $T_p/\mathbb{F}_p = F + I_p^*V$, and*

(i) *F can be identified with the inverse of the (“arithmetic”) Frobenius element φ_p of the Galois group $\text{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ acting on $\tilde{H}^1(M_n \otimes \overline{\mathbb{F}}_p, \text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Z}}_\ell))$;*

(ii) *F and V are transposes of one another, relative to the scalar product (3.20).*

(iii) *$FV = VF = p^{k+1}$.*

For the relation (i) between the geometric and arithmetic Frobenius, one is referred to the exposé of C. Houzel (SGA 5.XV). The composite VF is the composite of the homomorphisms obtained from the following maps:

$$\begin{array}{ccccccc} E & \leftarrow & E^{(p)} & \xrightarrow{F_E} & E & \xrightarrow{V_E} & E^{(p)} & \rightarrow & E \\ \downarrow & & & \searrow & \downarrow & \swarrow & & & \downarrow \\ M_n & & \xrightarrow{F} & & M_n & & \xrightarrow{F} & & M_n \\ & & & & VF = \text{Tr}_F \circ F_E^* \circ V_E^* \circ F^* & & & & \end{array}$$

The map $F_E^* V_E^* = (F_E V_E)^* = (p \cdot 1_E)^*$ acts by multiplication by p^k on $\text{Sym}^k R^1 f_{n*} \underline{\mathbb{Z}}_\ell$, so that $VF = p^k \cdot \text{Tr}_F \circ F^* = p^k \cdot p = p^{k+1}$ because $F: M_n \rightarrow M_n$ is of degree p .

By transport of structure, φ_p resects the scalar product (3.20) with values in $\mathbb{Q}_\ell(-k-1)$, a group on which φ_p acts by multiplication by p^{-k-1} . One has thus

$$(Fx, y) = p^{k+1}(\varphi_p Fx, \varphi_p y) = (x, p^{k+1} F^{-1}y) = (x, Vy). \quad \square$$

The following theorem, synonymous with (4.8), goes back to Eichler.

Theorem 4.9 (The Congruence Formula) *Let $K_{n,\ell}$ be the largest subextension of $\overline{\mathbb{Q}}$ that is unramified away from n and ℓ , let φ_p be a Frobenius element relative to p in $\text{Gal}(K_{n,\ell}/\mathbb{Q})$, let F be the endomorphism φ_p^{-1} of ${}_n W_\ell$, and let V be the transpose of F relative to the scalar product (3.20). Then,*

$$T_p = F + I_p^*V, \quad FV = p^{k+1}, \quad \text{and} \quad 1 - T_p X + p R_p X^2 = (1 - FX)(1 - I_p^* V X). \quad \square$$

5 Weil implies Ramanujan

If p is a prime number and X a scheme over $\mathbb{F}_p$, then one denotes by $\overline{\mathbb{F}}_p$ an algebraic closure of $\mathbb{F}_p$, by $F: X \rightarrow X$ the (“geometric”) Frobenius endomorphism, and one puts $\overline{X} = X \otimes \overline{\mathbb{F}}_p$; ℓ always denotes a prime number different from p .

By the “Weil conjectures,” one means to claim the following:

— *Let X be a projective smooth scheme over $\mathbb{F}_p$ and ℓ a prime number different from p . Then, the eigenvalues of the endomorphism F^* of $H^i(\overline{X}, \mathbb{Q}_\ell)$ are algebraic integers all of whose complex conjugates have absolute value $p^{i/2}$.*

With the hypotheses and notations of (4.9), one has (recall that $(p, n) = 1$):

Theorem 5.1 *If the Weil conjectures are true, then the absolute values of the endomorphism F of ${}^k_n W_\ell$ are algebraic integers (all of whose complex conjugates are) of absolute value $p^{k+1/2}$.*

Assume the Weil conjectures.

Lemma 5.2 (modulo Weil) *Let X be a smooth scheme over $\mathbb{F}_p$, which can be represented as an open in a smooth projective scheme X^* . Then, the absolute values of endomorphism F^* of $\tilde{H}^i(\overline{X}, \mathbb{Q}_\ell)$ are algebraic integers of absolute value $p^{i/2}$.*

The natural map of $H_c^i(\overline{X}, \mathbb{Q}_\ell)$ to $H^i(\overline{X}, \mathbb{Q}_\ell)$ can be factored through $H^i(\overline{X}^*, \mathbb{Q}_\ell)$:

$$H_c^i(\overline{X}, \mathbb{Q}_\ell) \rightarrow H^i(\overline{X}^*, \mathbb{Q}_\ell) \rightarrow H^i(\overline{X}, \mathbb{Q}_\ell)$$

so that as a $\text{Gal}(\overline{\mathbb{F}_p}/\mathbb{F}_p)$ -module, $\tilde{H}^i(\overline{X}, \mathbb{Q}_\ell)$ is the quotient of a subobject of $H^i(\overline{X}^*, \mathbb{Q}_\ell)$. $\square$

Lemma 5.3 (modulo Weil) *Let S be a smooth scheme over $\mathbb{F}_p$ and $f: A \rightarrow S$ an abelian scheme over S . Suppose that A can be represented as an open in a projective smooth scheme A^* over $\mathbb{F}_p$. Then, the geometric Frobenius F^* of $\tilde{H}^i(\overline{S}, R^j f_* \underline{\mathbb{Q}}_\ell)$ has eigenvalues that are algebraic integers of absolute value $p^{i+j/2}$.*

Let m be an integer > 1 , and consider the Leray spectral sequences,

$$\begin{aligned} E &: E_2^{ij} = H^i(\overline{S}, R^j f_* \underline{\mathbb{Q}}_\ell) \implies H^{i+j}(\overline{A}, \mathbb{Q}_\ell), \\ {}_c E &: {}_c E_2^{ij} = H_c^i(\overline{S}, R^j f_* \underline{\mathbb{Q}}_\ell) \implies H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell). \end{aligned}$$

The endomorphism of multiplication by m : $\psi_m = m \cdot 1_A$, defines endomorphisms of E and ${}_c E$ that fit into a commutative diagram:

$$\begin{array}{ccc} {}_c E & \rightarrow & E \\ \psi_m^* \downarrow & & \psi_m^* \downarrow \\ {}_c E & \rightarrow & E \end{array}.$$

ψ_m^* acts on $R^j f_* \underline{\mathbb{Q}}_\ell$ by multiplication by m^j , so that ψ_m^* acts on E by multiplication by m^j on the terms ${}_c E_r^{ij}$ and E_r^{ij} of ${}_c E$ and E . The maps d_r ($r \geq 2$) commute with ψ_m^* , and send E_r^{ij} (resp. ${}_c E_r^{ij}$) to $E_r^{i'j'}$ (resp. ${}_c E_r^{i'j'}$) with $j \neq j'$. These are nonzero, and E_2^{ij} (resp. ${}_c E_2^{ij}$) can be identified with a subspace of $H^{i+j}(\overline{A}, \mathbb{Q}_\ell)$ (resp. of $H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell)$) where $\psi_m^* = m^j$. Consequently, $\tilde{H}^i(\overline{S}, R^j f_* \underline{\mathbb{Q}}_\ell)$ is identified, as a Galois module, with a subspace of $\tilde{H}^{i+j}(\overline{A}, \mathbb{Q}_\ell)$ where $\psi_m^* = m^j$ and one applies (5.2). The trick used here is due to Lieberman. $\square$

Let $f_n: E \rightarrow M_n \otimes \mathbb{F}_p$ be the universal elliptic curve over $M_n \otimes \mathbb{F}_p$, let $f_{n,k}: E_k \rightarrow M_n \otimes \mathbb{F}_p$ be its k -fold fiber product with itself. The Kunneth formula shows that the $\mathbb{Q}_\ell$ -sheaf $R^k f_{n,k*} \underline{\mathbb{Q}}_\ell$ admits as a direct factor the k -fold tensor power of $R^1 f_{n*} \underline{\mathbb{Q}}_\ell$; the latter in turn contains as a direct factor the $\mathbb{Q}_\ell$ -sheaf $\text{Sym}^k(R^1 f_{n*} \underline{\mathbb{Q}}_\ell)$. Theorem 5.1 results thus from (5.3) and

Lemma 5.4 *The scheme E_k is an open in a smooth projective scheme E_k^* over $\mathbb{F}_p$.*

Let E^* be the Néron minimal model of E over $M_n^* \otimes \mathbb{F}_p$ (4.1). The scheme E^* is smooth and projective over $\mathbb{F}_p$. Since $n \geq 3$ and the points of order n of E form a trivial covering of $M_n \otimes \mathbb{F}_p$, the Néron model is “semistable” (case a or b_m in Néron’s classification). In particular, the projection $f_n: E^* \rightarrow M_n^*$ has only a finite number of nonsmooth points, and at these points, f_n is nondegenerate (presents an ordinary quadratic singularity).

Let E_k^{**} be the k -fold fiber product of E^* over M_n^* . To prove (5.4), it suffices to resolve the singularities of E_k^{**} without touching the open E_k . We prove first:

Lemma 5.5 *Let V be the subvariety of affine space over a field k (coordinates $(X_i)_{0 \leq i \leq r}$, $(Y_i)_{0 \leq i \leq r}$, $(T_i)_{1 \leq i \leq s}$) with equation*

$$X_0 Y_0 = X_1 Y_1 = \cdots = X_r Y_r.$$

Let $\mathfrak{m}$ be the ideal of $\mathcal{O}_V$ generated by the monomials obtained from the monomial $\prod_{i=0}^r X_i^i$ by a permutation of the coordinates which respects the set of pairs $\{X_i, Y_i\}$ ($0 \leq i \leq r$). Then, $\mathfrak{m} = \mathcal{O}_V$ away from the singular locus of V , and the variety $\tilde{V}$ is obtained from V by blowing up $\mathfrak{m}$ is smooth over k .

The singular locus is the place where the four coordinates X_i, Y_i, X_j, Y_j ($i \neq j$) vanish. The open affine of $\tilde{V}$ defined by the element $\prod_{i=0}^r X_i^i$ of the ideal $\mathfrak{m}$ of blowing up is the spectrum of the regular ring

$$k[Y_0/X_1, X_0/X_1, X_1/X_2, \dots, X_{r-1}/X_r, X_r, T_1, \dots, T_s]$$

(for the proof, note that $X_i/X_{i+1} = Y_{i+1}/Y_i$), and 5.5 results. $\square$

One shows still that, locally for the étale topology, the singularities of E_k^{**} are isomorphic to those of V (for $r = k - 1$), and that this allows the definition on E_k^{**} of an ideal $\mathfrak{m}$ analogous to the ideal $\mathfrak{m}$ of (5.5). Blowing up this ideal, one obtains E_k^* . $\square$

An approximation to the following theorem has been proven by Ihara [2]:

Theorem 5.6 *The Weil conjectures imply the Ramanujan conjecture.*

We note first that (5.1) is true for $n = 1$, because ${}^k_1 W_\ell$ is the Galois submodule of ${}^k_m W_\ell$ invariant under $GL_2(\mathbb{Z}/(m))$. On ${}^k_1 W_\ell$, I_p^* induces the identity, and (4.8) reduces to

$$1 - T_p X + p^{k+1} X^2 = (1 - F X)(1 - V X).$$

The endomorphisms F and V are transposes of each other, so that

$$\det(1 - F X; {}^k_1 W_\ell) = \det(1 - V X; {}^k_1 W_\ell).$$

The action of T_p on ${}^k_1 W_\ell$ is induced by its action on ${}^k_1 W$, and is compatible with the decomposition of ${}^k_1 W \otimes \mathbb{C}$ into the sum of the space S_{k+2} of cusp forms relative to $SL_2(\mathbb{Z})$ of weight $k + 2$, and of the complex conjugate space. From the Hermitian property for T_p (for the Peterson inner product), and from (3.19), one deduces then that

$$\det(1 - T_p X + p^{k+1} X^2; {}^k_1 W_\ell) = \det(1 - T_p X + p^{k+1} X^2; S_{k+2})^2,$$

and

$$\det(1 - T_p X + p^{k+1} X^2; S_{k+2})^2 = \det(1 - FX; {}_1^k W_\ell)^2,$$

so that

$$\det(1 - T_p X + p^{k+1} X^2; S_{k+2}) = \det(1 - FX; {}_1^k W_\ell). \quad (5.7)$$

Resume the notations of n° 1 and make $k = 10$. By vurtue of Hecke's theory and of (3.19), (5.7) can be rewritten

$$H_p(X) = \det(1 - FX; {}_{10}^1 W_\ell)$$

and one applies (5.1). $\square$

One verifies in the same way that the Weil conjectures imply Pertersson's generalization of the Ramanujan conjecture.

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Initials:

EGA: *Eléments de géométrie algébrique*, by A. Grothendieck and J. Dieudonné, Publ. Math. I.H.E.S.

SGA: *Séminaire de géométrie algébrique du Bois-Marie*, Mimeographed notes by I.H.E.S., in preparation by North-Holland.